

MailSec TLS Executive Security Audit

Capture: smtp_vulnerable_sample.pcap | SHA-256: 4dc438ee4e2df992b386a2ae...

Audit Generated: 2026-09-28T14:27:06 UTC | Engine: SCAPY

SECURITY GRADE	DETERMINISTIC RISK SCORE	AI ANOMALY RISK INDEX
F CRITICAL RISK	100 4 Findings (2 Critical)	75 2 Flagged Anomalies

Protocol & Cryptographic Telemetry

Metric	Value	Metric	Value
Total Sessions	2	STARTTLS Upgrades	1 (50%)
SMTP Streams	2	TLS Encrypted	1
IMAP / POP3 Streams	0 / 0	Cleartext Insecure	1
PFS Adoption (ECDHE)	0	TLS 1.3 / 1.2 / Legacy	0 / 0 / 1

Security Vulnerability Findings (4)

Severity	Rule ID	Affected Entity	Description & Remediation
CRITICAL	RULE_CLEARTEXT_EMAIL	Stream #1 (192.168.1.110 -> 192.168.1.25:25)	Cleartext SMTP Session Transmitted Without Encryption The SMTP session was transmitted completely unencrypted over the wire without STARTTLS. Credentials, message envelope headers, and email con... Remediation: Enforce mandatory TLS encryption on mail servers (implicit TLS on ports 465/993/995 or set 'smtpd_tl...
CRITICAL	RULE_CIPHER_INSECURE	Stream #2 (TLS_RSA_WITH_3DES_EDE_CBC_SHA)	Insecure or Broken Cipher Suite Negotiated (TLS_RSA_WITH_3DES_EDE_CBC_SHA) The session negotiated 'TLS_RSA_WITH_3DES_EDE_CBC_SHA', which relies on obsolete cryptographic primitives (e.g., 3DES vulnerable to 64-bit S... Remediation: Eliminate legacy 3DES, RC4, and DES ciphers from the server cipher suite list. Configure modern AEAD...
HIGH	RULE_DEPRECATED_TLS_VERSION	Stream #2 (TLS 1.0)	Deprecated TLS Protocol Version Negotiated (TLS 1.0) The session negotiated TLS 1.0, which is formally deprecated by RFC 8996 and NIST. Legacy TLS versions lack resistance to known cryptographi... Remediation: Disable TLS 1.0 and 1.1 in mail server configuration. Configure 'smtpd_tls_protocols = !SSLv2, !SSLv...

MEDIUM	RULE_NO_FORWARD_SECRECY	Stream #2 (RSA)	Static RSA Key Exchange Lacks Perfect Forward Secrecy The session negotiated static RSA key exchange without ephemeral Diffie-Hellman (ECDHE / DHE). If the server's private key is compromised at... Remediation: Configure your mail server cipher suites to prefer ECDHE key exchange (e.g. using X25519 or secp256r...
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AI / Machine Learning Anomalies (2)

Stream	Protocol	Endpoints	Anomaly Score	Primary Reasons
#1	SMTP	192.168.1.110:49152 -> 192.168.1.25:25	75.0%	• Completely cleartext SMTP communication without encryption on mail port 25
#2	SMTP	192.168.1.112:50210 -> 192.168.1.80:587	775.0%	• Negotiated deprecated legacy TLS version (TLS 1.0) • Insecure cipher suite negotiated (TLS_RSA_WITH_3DES_EDE_CBC_SHA) vulnerable to cryptanalytic collision attacks • Static RSA key exchange lacks Perfect Forward Secrecy (PFS), susceptible to retroactive decryption • Client offered high proportion of weak/non-PFS ciphers (100% of offered suites)

X.509 Certificate Inventory (0)

No X.509 certificates extracted from capture.